

Library Management System — Project Report

Course: Spring Boot Web Application Development **Institution:** BITS Pilani **GitHub:** [@Manasvi-247](#) · [library-management-springboot](#)

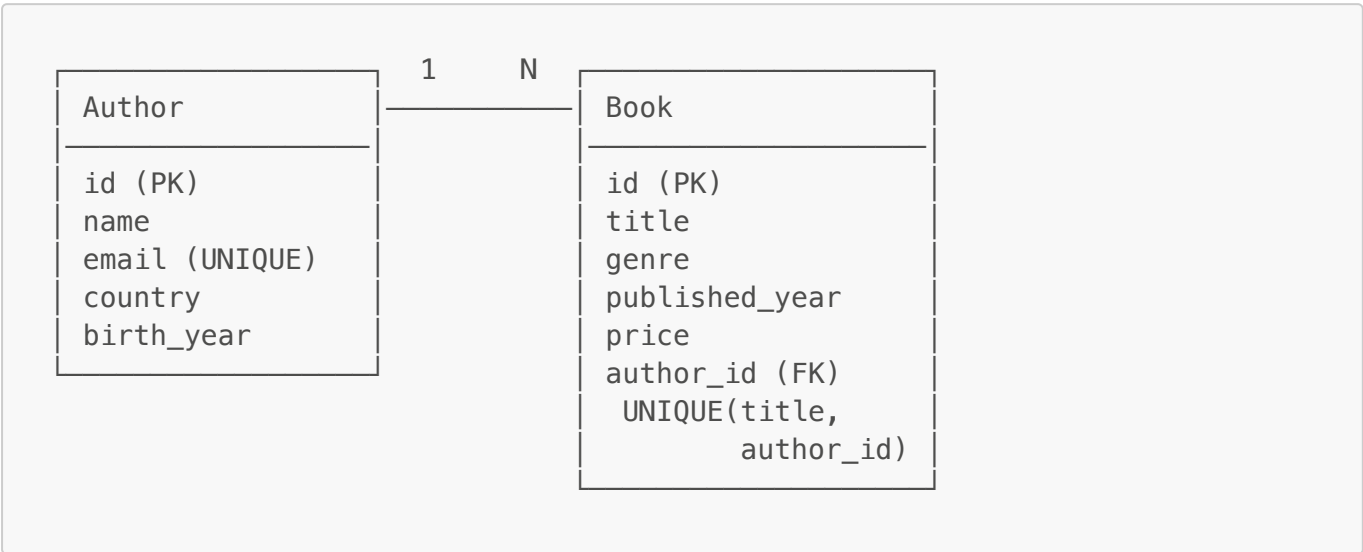
1. Overview

Inkwell is a Spring Boot reference application that manages two related entities — **Author** and **Book** — through a web UI built with JSP. The application demonstrates:

- JPA entity modelling with a One-to-Many relationship.
- Repository, service, and controller layering with clear separation of concerns.
- Three CRUD operations per the assignment (**Create, Read, Update**).
- A custom **INNER JOIN** JPQL query projecting into a DTO.
- Form validation, integrity-violation handling, and a polished JSP UI.
- Unit tests for the repository and service layers.

2. Entity-Relationship Design

Two entities, related as **Author 1 — N Book**:



Constraints (also drive the integrity-violation demo)

Constraint	Why
<code>Author.email</code> UNIQUE	Prevents two authors sharing an email.
<code>Book(title, author_id)</code> UNIQUE composite	Same author cannot have two books with the same title.
<code>Book.author_id</code> NOT NULL + FK	Every book must reference an existing author.
Bean Validation on form fields	Client-side rejection before the DB is touched.

JPA mapping highlights

- `Author.books` — `@OneToMany(mappedBy = "author", cascade = ALL, orphanRemoval = true)`
- `Book.author` — `@ManyToOne(fetch = LAZY, optional = false) + @JoinColumn(name = "author_id")`
- Bean Validation: `@NotBlank`, `@Email`, `@Min`, `@Positive`, `@NotNull`.

File references: `Author.java`, `Book.java`

3. Layered Architecture

```
JSP views — Controllers — Services — Repositories — Database
                        |
                        └─ @ControllerAdvice
(GlobalExceptionHandler)
```

Layer	Responsibility
Controller	HTTP request/response, model binding, validation, flash messages.
Service	Transactional business logic, domain exceptions.
Repository	Spring Data JPA — derived methods + custom JPQL.
View	JSP + JSTL/EL with form-binding tags.

4. Implementation Details

4.1 Populating the database

`config/DataSeeder.java` is a `CommandLineRunner` that inserts **10 authors and 10 books** on startup if the tables are empty. This guarantees the grader sees a populated database the moment the app starts.

```
@Component
public class DataSeeder implements CommandLineRunner {
    public void run(String... args) {
        if (authorRepository.count() > 0) return;
        // 10 authors + 10 books inserted via saveAll(...)
    }
}
```

The screenshot shows the H2 console interface. The left sidebar displays the database schema with tables: AUTHORS, BOOKS, INFORMATION_SCHEMA, and Users. The main area shows the SQL statement: `SELECT * FROM AUTHORS;` and `SELECT * FROM BOOKS;`. Below the query, the results for both tables are displayed.

SELECT * FROM AUTHORS;

ID	BIRTH_YEAR	COUNTRY	EMAIL	NAME
1	1974	India	chetan@inkwell.in	Chetan Bhagat
2	1967	India / USA	jhumpa@inkwell.in	Jhumpa Lahiri
3	1961	India	arundhati@inkwell.in	Arundhati Roy
4	1947	United Kingdom	rushdie@inkwell.uk	Salman Rushdie
5	1949	Japan	murakami@inkwell.jp	Haruki Murakami
6	1965	United Kingdom	jkr@inkwell.uk	J.K. Rowling
7	1948	United States	grm@inkwell.us	George R. R. Martin
8	1939	Canada	atwood@inkwell.ca	Margaret Atwood
9	1976	Israel	harari@inkwell.il	Yuval Noah Harari
10	1947	Brazil	coelho@inkwell.br	Paulo Coelho

(10 rows, 0 ms)

SELECT * FROM BOOKS;

ID	GENRE	PRICE	PUBLISHED_YEAR	TITLE	AUTHOR_ID
1	Fiction	199.00	2004	Five Point Someone	1
2	Literary	349.00	2003	The Namesake	2
3	Literary	450.00	1997	The God of Small Things	3
4	Magical Realism	499.00	1981	Midnight's Children	4
5	Literary	550.00	1987	Norwegian Wood	5
6	Fantasy	399.00	1997	Harry Potter & the Sorcerer's Stone	6
7	Fantasy	699.00	1996	A Game of Thrones	7
8	Dystopian	525.00	1985	The Handmaid's Tale	8
9	Non-Fiction	799.00	2011	Sapiens	9
10	Fiction	299.00	1988	The Alchemist	10

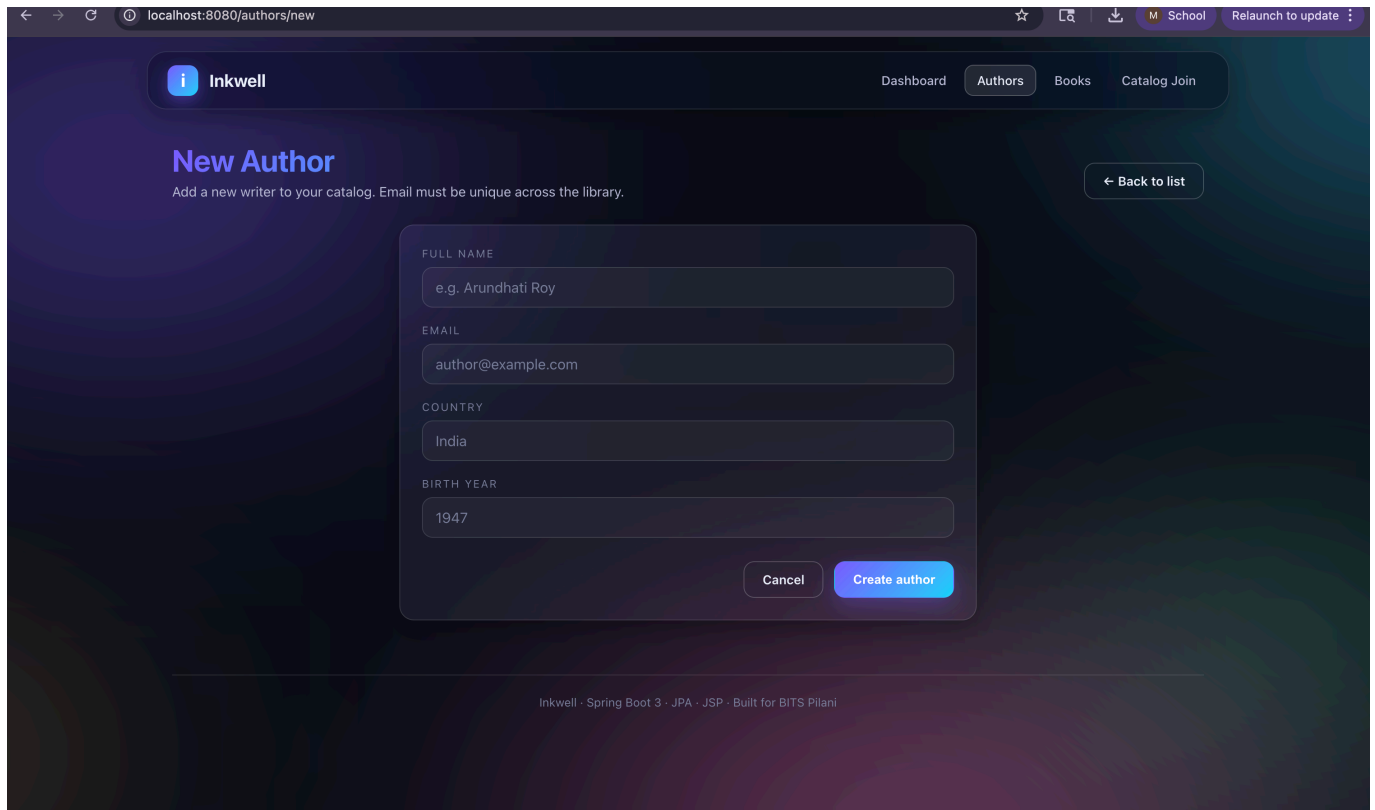
(10 rows, 1 ms)

Figure 1 — H2 console: both **AUTHORS** and **BOOKS** tables populated with the 10 + 10 sample rows from **DataSeeder**.

4.2 Create operation

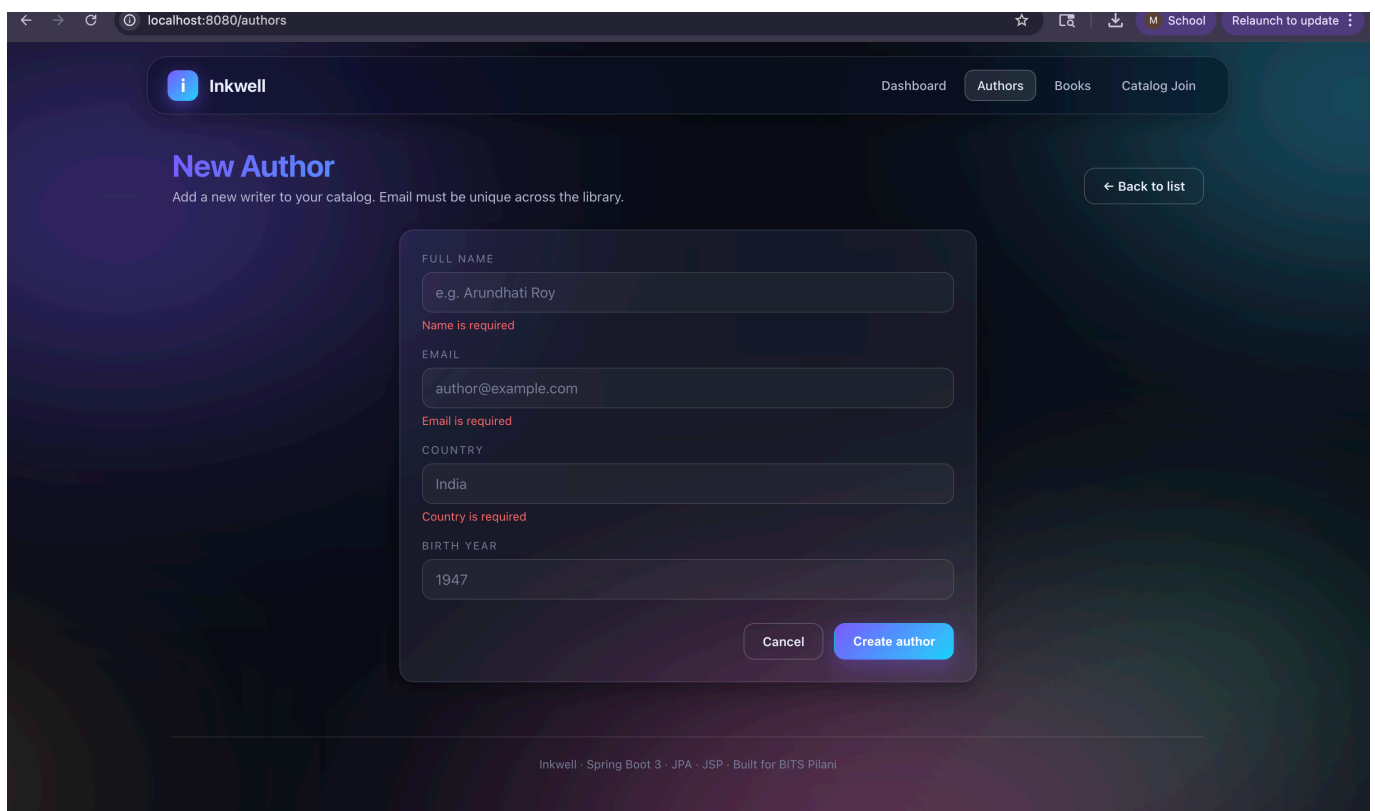
- JSP form: `authors/form.jsp`, `books/form.jsp`
- Controller: `POST /authors` and `POST /books`
- `@Valid` triggers Bean Validation; errors are re-rendered into the same form via `BindingResult`.
- Successful creates redirect with a flash message ("Author 'X' created successfully").
- Duplicate email or duplicate (`title`, `author`) triggers `DataIntegrityViolationException`, caught by the global handler and rendered through `error.jsp`.

```
@PostMapping
public String create(@Valid @ModelAttribute("author") Author author,
                    BindingResult bindingResult, ...) {
    if (bindingResult.hasErrors()) return "authors/form";
    Author saved = authorService.create(author);
    redirect.addFlashAttribute("flashSuccess", "Author '" + saved.getName()
+ "' created.");
    return "redirect:/authors";
}
```



The screenshot shows a web browser at `localhost:8080/authors/new`. The application header includes the 'Inkwell' logo and navigation links for 'Dashboard', 'Authors', 'Books', and 'Catalog Join'. The main heading is 'New Author' with a subtext: 'Add a new writer to your catalog. Email must be unique across the library.' A 'Back to list' button is in the top right. The form itself has four input fields: 'FULL NAME' (with placeholder 'e.g. Arundhati Roy'), 'EMAIL' (with placeholder 'author@example.com'), 'COUNTRY' (with placeholder 'India'), and 'BIRTH YEAR' (with placeholder '1947'). At the bottom of the form are 'Cancel' and 'Create author' buttons. The footer text reads 'Inkwell · Spring Boot 3 · JPA · JSP · Built for BITS Pilani'.

Figure 2 — `GET /authors/new`: empty form rendered by `authors/form.jsp`.



This screenshot shows the same 'New Author' form, but now with validation errors. Red text messages are displayed below the input fields: 'Name is required' under the Full Name field, 'Email is required' under the Email field, and 'Country is required' under the Country field. The Birth Year field remains valid. The 'Create author' button is still highlighted in blue. The rest of the page layout, including the header and footer, is identical to the previous figure.

Figure 3 — Bean Validation errors surfaced via `BindingResult` and Spring's `<form:errors>` tag — the form re-renders with red error messages under each field.

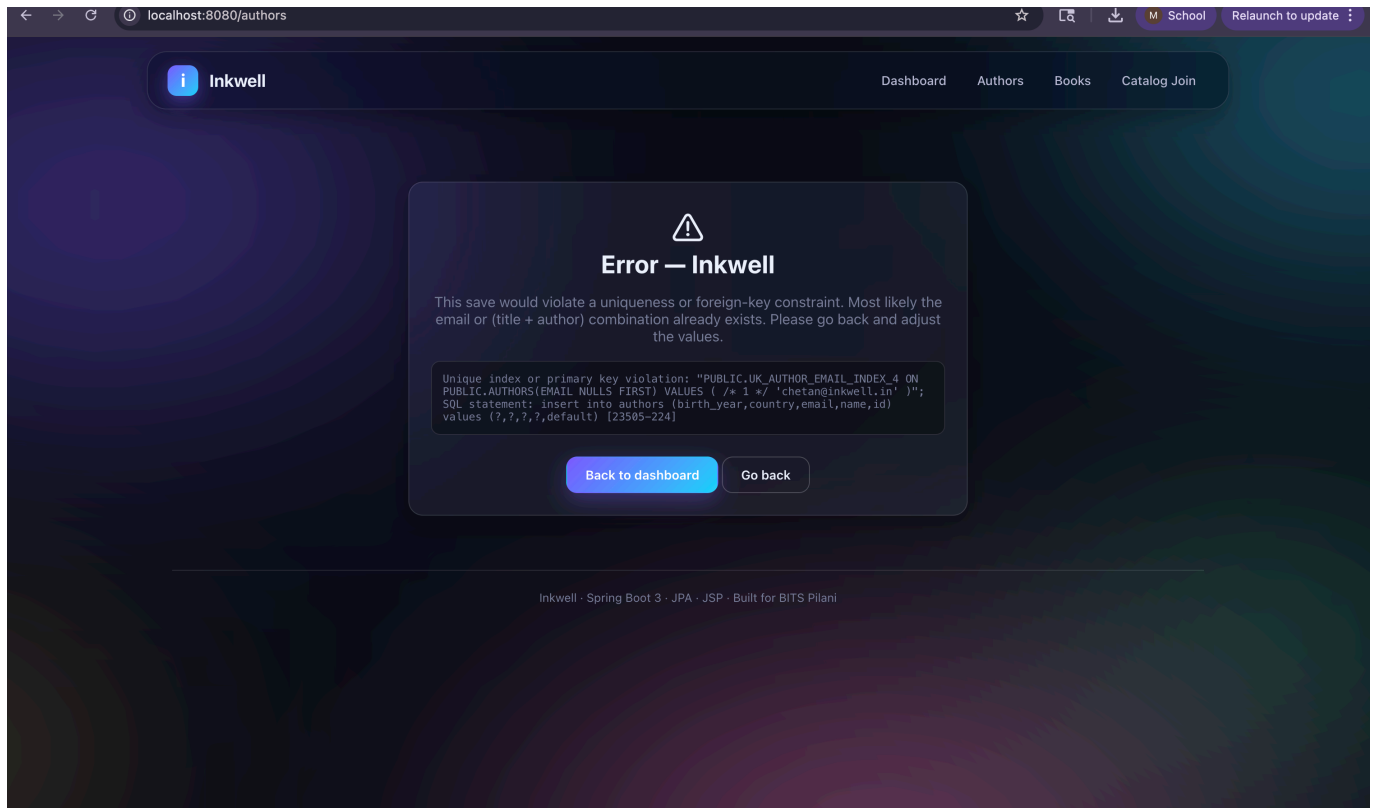


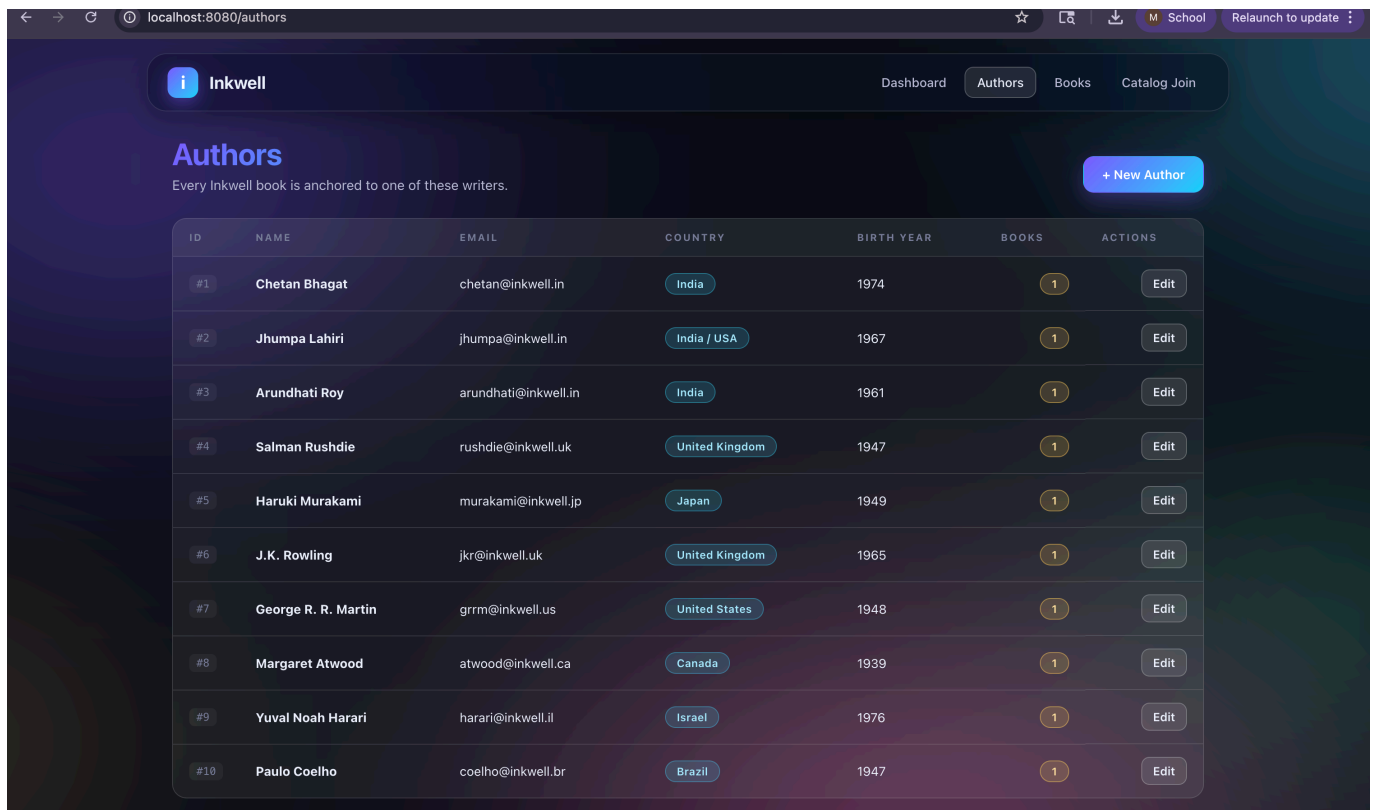
Figure 4 — Duplicate email triggers *DataIntegrityViolationException*, caught by *GlobalExceptionHandler* and rendered as a styled 409 page with the underlying constraint name.

4.3 Read operation

- List views at */authors*, */books*.
- The **inner-join** projection lives at */books/joined* and uses this JPQL:

```
@Query("SELECT new com.bits.library.dto.AuthorBookView(" +
    " a.name, a.country, b.title, b.genre, b.publishedYear, b.price) "
+
    "FROM Book b INNER JOIN b.author a " +
    "ORDER BY a.name ASC, b.publishedYear DESC")
List<AuthorBookView> findAllAuthorsWithBooks();
```

The view layer never touches lazy associations because the query returns a flat *AuthorBookView* DTO. A second variant (*findByGenreJoined*) supports filtering by genre — wired up to a small filter bar in the joined view.



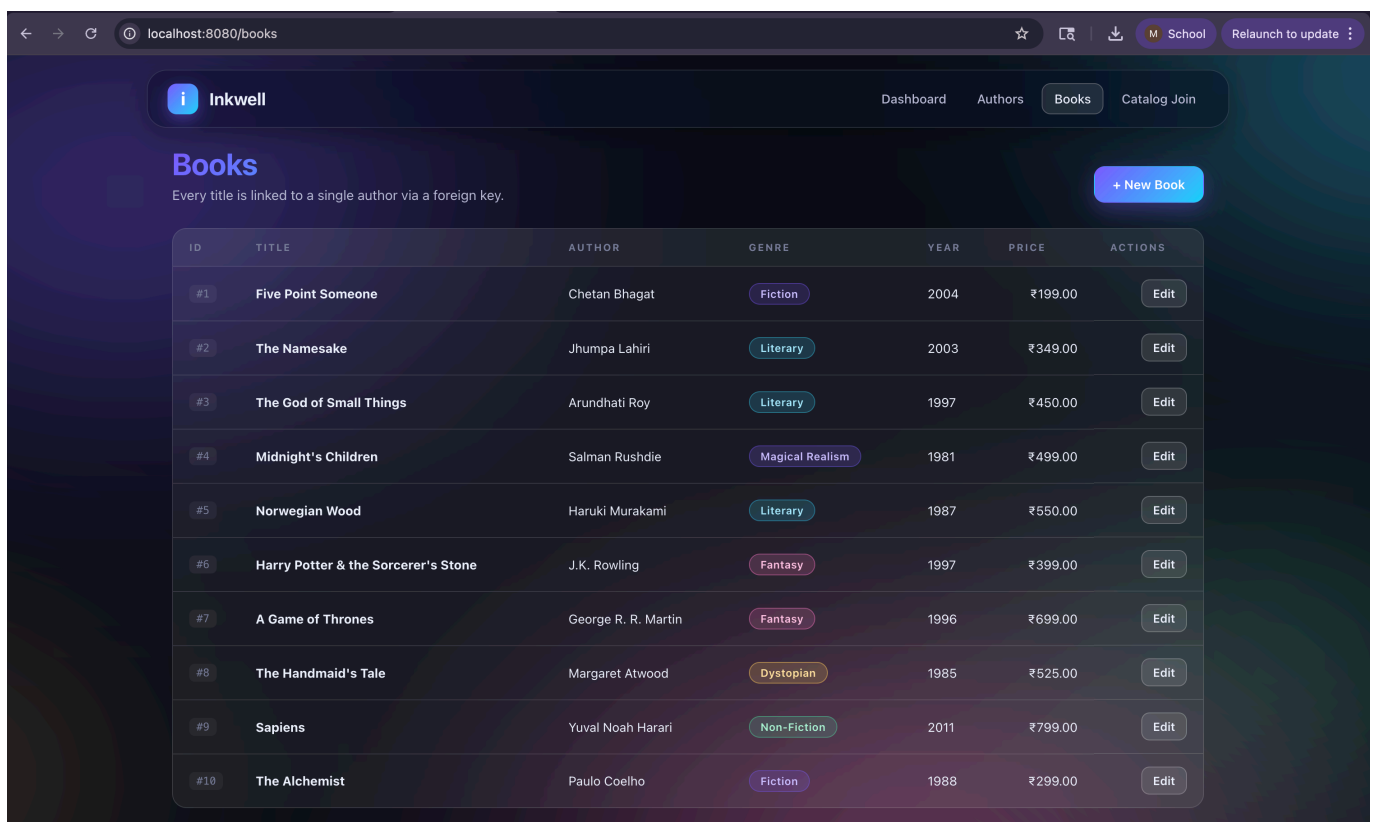
Inkwell Dashboard Authors Books Catalog Join

Authors

Every Inkwell book is anchored to one of these writers. [+ New Author](#)

ID	NAME	EMAIL	COUNTRY	BIRTH YEAR	BOOKS	ACTIONS
#1	Chetan Bhagat	chetan@inkwell.in	India	1974	1	Edit
#2	Jhumpa Lahiri	jhumpa@inkwell.in	India / USA	1967	1	Edit
#3	Arundhati Roy	arundhati@inkwell.in	India	1961	1	Edit
#4	Salman Rushdie	rushdie@inkwell.uk	United Kingdom	1947	1	Edit
#5	Haruki Murakami	murakami@inkwell.jp	Japan	1949	1	Edit
#6	J.K. Rowling	jkr@inkwell.uk	United Kingdom	1965	1	Edit
#7	George R. R. Martin	grrm@inkwell.us	United States	1948	1	Edit
#8	Margaret Atwood	atwood@inkwell.ca	Canada	1939	1	Edit
#9	Yuval Noah Harari	harari@inkwell.il	Israel	1976	1	Edit
#10	Paulo Coelho	coelho@inkwell.br	Brazil	1947	1	Edit

Figure 5 — `GET /authors`: all 10 authors rendered with country pills, book-count badges, and edit actions.



Inkwell Dashboard Authors Books Catalog Join

Books

Every title is linked to a single author via a foreign key. [+ New Book](#)

ID	TITLE	AUTHOR	GENRE	YEAR	PRICE	ACTIONS
#1	Five Point Someone	Chetan Bhagat	Fiction	2004	₹199.00	Edit
#2	The Namesake	Jhumpa Lahiri	Literary	2003	₹349.00	Edit
#3	The God of Small Things	Arundhati Roy	Literary	1997	₹450.00	Edit
#4	Midnight's Children	Salman Rushdie	Magical Realism	1981	₹499.00	Edit
#5	Norwegian Wood	Haruki Murakami	Literary	1987	₹550.00	Edit
#6	Harry Potter & the Sorcerer's Stone	J.K. Rowling	Fantasy	1997	₹399.00	Edit
#7	A Game of Thrones	George R. R. Martin	Fantasy	1996	₹699.00	Edit
#8	The Handmaid's Tale	Margaret Atwood	Dystopian	1985	₹525.00	Edit
#9	Sapiens	Yuval Noah Harari	Non-Fiction	2011	₹799.00	Edit
#10	The Alchemist	Paulo Coelho	Fiction	1988	₹299.00	Edit

Figure 6 — `GET /books`: all 10 books with genre pills (Fiction / Literary / Magical Realism / Fantasy / Dystopian / Non-Fiction) and prices.

Catalog Join
Result of an **INNER JOIN** between Book and Author, projected through a JPQL constructor expression into AuthorBookView.

Filter by genre (e.g. Ficti) **Apply**

```
SELECT new com.bits.library.dto.AuthorBookView(
    a.name, a.country, b.title, b.genre, b.publishedYear, b.price)
FROM Book b
INNER JOIN a:author a
ORDER BY a.name ASC, b.publishedYear DESC
```

AUTHOR × BOOK	COUNTRY	GENRE	YEAR	PRICE
Arundhati Roy > The God of Small Things	India	Literary	1997	₹450.00
Chetan Bhagat > Five Point Someone	India	Fiction	2004	₹199.00
George R. R. Martin > A Game of Thrones	United States	Fantasy	1996	₹699.00
Haruki Murakami > Norwegian Wood	Japan	Literary	1987	₹550.00
J.K. Rowling > Harry Potter & the Sorcerer's Stone	United Kingdom	Fantasy	1997	₹399.00
Jhumpa Lahiri > The Namesake	India / USA	Literary	2003	₹349.00
Margaret Atwood > The Handmaid's Tale	Canada	Dystopian	1985	₹525.00
Paulo Coelho > The Alchemist	Brazil	Fiction	1988	₹299.00
Salman Rushdie > Midnight's Children	United Kingdom	Magical Realism	1981	₹499.00
Yuval Noah Harari > Sapiens	Israel	Non-Fiction	2011	₹799.00

Figure 7 — `GET /books/joined`: result of the JPQL **INNER JOIN**, projected into **AuthorBookView**. The exact query is shown above the table for transparency.

Catalog Join
Result of an **INNER JOIN** between Book and Author, projected through a JPQL constructor expression into AuthorBookView.

Fantasy **Apply** **Clear**

```
SELECT new com.bits.library.dto.AuthorBookView(
    a.name, a.country, b.title, b.genre, b.publishedYear, b.price)
FROM Book b
INNER JOIN a:author a
WHERE LOWER(b.genre) = LOWER('Fantasy')
ORDER BY a.name ASC, b.publishedYear DESC
```

AUTHOR × BOOK	COUNTRY	GENRE	YEAR	PRICE
J.K. Rowling > Harry Potter & the Sorcerer's Stone	United Kingdom	Fantasy	1997	₹399.00
George R. R. Martin > A Game of Thrones	United States	Fantasy	1996	₹699.00

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Figure 8 — `GET /books/joined?genre=Fantasy`: same query with an added **WHERE LOWER(b.genre) = LOWER(?)** clause — returns just the two Fantasy titles.

4.4 Update operation

- Edit forms at `/authors/{id}/edit` and `/books/{id}/edit`.

- Submitting `POST /authors/{id}` or `POST /books/{id}` calls `authorService.update(id, ...)` / `bookService.update(id, ...)`.
- The service fetches the existing entity, copies in the new field values, and re-saves — keeping the JPA-managed identity stable.

localhost:8080/authors/1/edit

Inkwell Dashboard Authors Books Catalog Join

Edit Author

Update existing author details. Email must remain unique.

[← Back to list](#)

FULL NAME
Chetan Bhagat

EMAIL
chetan@inkwell.in

COUNTRY
India

BIRTH YEAR
1974

Cancel Save changes

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Figure 9 — `GET /authors/1/edit`: form pre-populated with the existing record (Chetan Bhagat / India / 1974).

localhost:8080/authors

Inkwell Dashboard Authors Books Catalog Join

Author 'Chetan Bhagat' updated successfully.

Authors

Every Inkwell book is anchored to one of these writers.

[+ New Author](#)

ID	NAME	EMAIL	COUNTRY	BIRTH YEAR	BOOKS	ACTIONS
#1	Chetan Bhagat	chetan@inkwell.in	Canada	1974	1	Edit
#2	Jhumpa Lahiri	jhumpa@inkwell.in	India / USA	1967	1	Edit
#3	Arundhati Roy	arundhati@inkwell.in	India	1961	1	Edit
#4	Salman Rushdie	rushdie@inkwell.uk	United Kingdom	1947	1	Edit
#5	Haruki Murakami	murakami@inkwell.jp	Japan	1949	1	Edit
#6	J.K. Rowling	jkr@inkwell.uk	United Kingdom	1965	1	Edit
#7	George R. R. Martin	grrm@inkwell.us	United States	1948	1	Edit
#8	Margaret Atwood	atwood@inkwell.ca	Canada	1939	1	Edit
#9	Yuval Noah Harari	harari@inkwell.il	Israel	1976	1	Edit

Figure 9b — After submission, the redirect renders `flashSuccess` ("Author 'Chetan Bhagat' updated

successfully") and the country pill now reads **Canada** — proving the update persisted through the service → repository → DB chain.

4.5 Exception handling

`GlobalExceptionHandler` (`@ControllerAdvice`) handles:

Exception	HTTP	User message
<code>DataIntegrityViolationException</code>	409	"Duplicate or invalid data — uniqueness/foreign-key broke."
<code>AuthorNotFoundException</code> / <code>Book...</code>	404	"Record not found."
Any other <code>Exception</code>	500	"Something went wrong" + cause detail.

All errors render through a single styled `error.jsp` page.

5. View Layer & UI

The UI is built on a small custom design system using only CSS (no framework):

- Animated mesh-gradient background with two overlapping radial layers.
- Glassmorphism cards (`backdrop-filter: blur`).
- Coloured "pill" badges for genre and country.
- Sticky frosted navbar with active-route highlighting.
- Responsive table that collapses into stacked rows on mobile.
- Flash messages, validation errors, and a friendly error page.

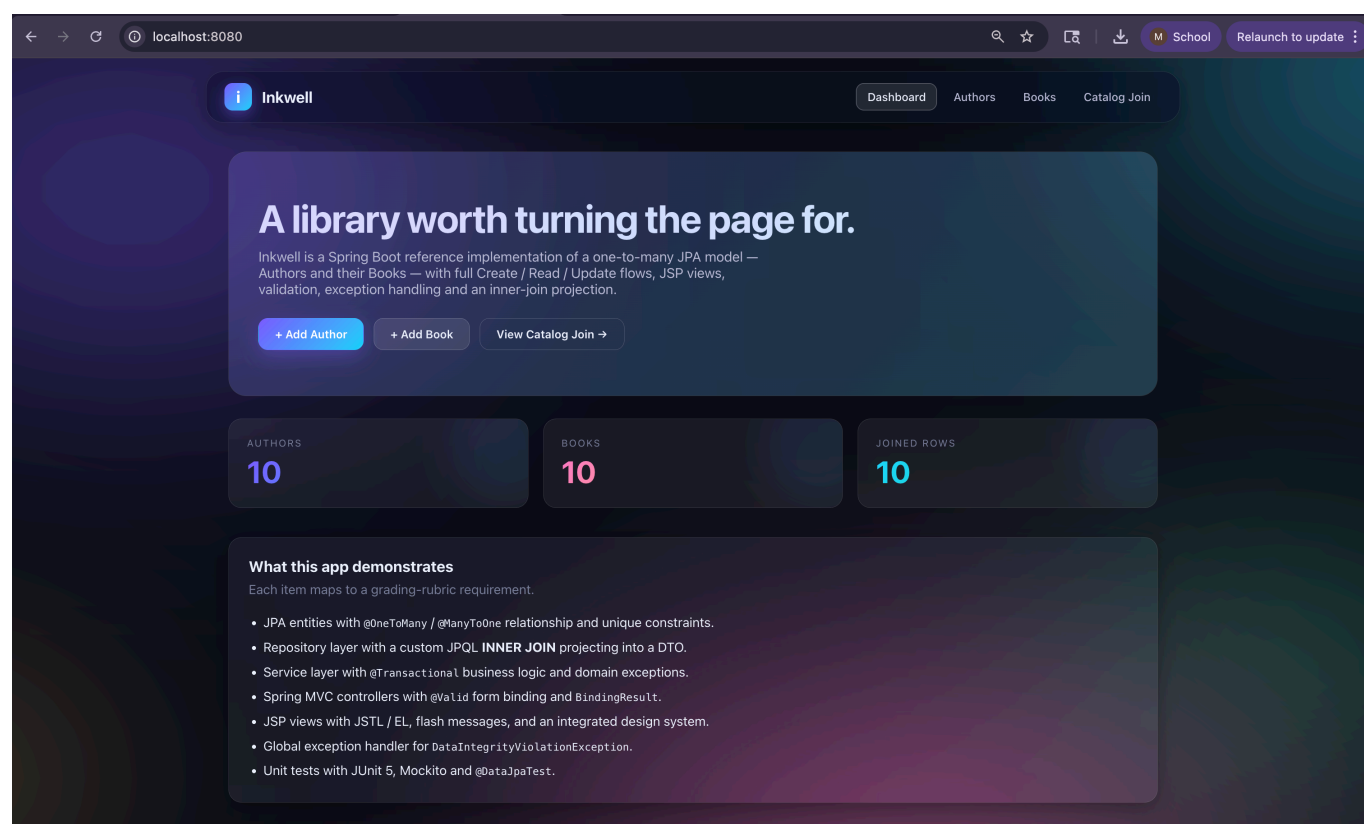


Figure 10 — `GET /`: dashboard hero with three live stat cards (Authors / Books / Joined Rows) plus a

feature-checklist card.

6. Testing & Validation

Test class	Type	What it proves
<code>BookRepositoryTest</code>	<code>@DataJpaTest</code>	Inner-join returns correct rows, excludes authors with no books, genre filter is case-insensitive, unique-email + unique-(title, author) constraints throw <code>DataIntegrityViolationException</code> .
<code>AuthorServiceTest</code>	Mockito	<code>findAll</code> , <code>findById</code> (success + miss), <code>create</code> , <code>update</code> field copy.
<code>BookServiceTest</code>	Mockito	Author resolution on create, field copy + author reassignment on update.
<code>LibraryApplicationTests</code>	<code>@SpringBootTest</code>	Full Spring context wires up cleanly.

Run with:

```
mvn test
```

```

> ▾ TERMINAL
insert
into
  books
(author_id, genre, price, published_year, title, id)
values
  (?, ?, ?, ?, ?, default)
2026-05-03T01:15:44.632+05:30 DEBUG 22463 --- [          main] org.hibernate.SQL      :
insert
into
  books
(author_id, genre, price, published_year, title, id)
values
  (?, ?, ?, ?, ?, default)
2026-05-03T01:15:44.633+05:30 DEBUG 22463 --- [          main] org.hibernate.SQL      :
insert
into
  books
(author_id, genre, price, published_year, title, id)
values
  (?, ?, ?, ?, ?, default)
2026-05-03T01:15:44.633+05:30 DEBUG 22463 --- [          main] org.hibernate.SQL      :
insert
into
  books
(author_id, genre, price, published_year, title, id)
values
  (?, ?, ?, ?, ?, default)
2026-05-03T01:15:44.634+05:30 INFO 22463 --- [          main] com.bits.library.config.DataSeeder      : Seeded 10 authors
and 10 books.
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.456 s -- in com.bits.library.LibraryApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.898 s
[INFO] Finished at: 2026-05-03T01:15:44+05:30
[INFO]
manasvi@Manasvis-MacBook-Pro bits %
  
```

Figure 11 — `mvn test`: all 14 unit tests pass across `BookRepositoryTest` (5), `BookServiceTest` (3), `AuthorServiceTest` (5), and `LibraryApplicationTests` (1). BUILD SUCCESS.

7. Running the App

See [README.md](#) for full setup.

Clone & run (H2 — zero setup):

```
git clone https://github.com/Manasvi-247/library-management-springboot.git
cd library-management-springboot
mvn spring-boot:run -Dspring-boot.run.profiles=h2
```

Then visit <http://localhost:8080/>.

MySQL profile (default):

```
mvn spring-boot:run
```

The schema `library_db` is auto-created via `createDatabaseIfNotExist=true`. Defaults to `root/root` — override in `src/main/resources/application-mysql.properties`.

Run tests:

```
mvn test
```

8. Challenges & Resolutions

Challenge	Resolution
JSP + Spring Boot 3 (Jakarta) JSTL imports	Used <code>jakarta.tags.core</code> URI and the matching Glassfish <code>jakarta.servlet.jsp.jstl</code> 3.0.x runtime — <code>javax.*</code> is dead.
<code>LazyInitializationException</code> when rendering <code>\${author.books}</code> in JSP	Two paths: keep <code>spring.jpa.open-in-view=true</code> , or project the joined data into a flat DTO. We use the DTO for the join page.
Surfacing integrity violations as friendly UI errors	<code>@ControllerAdvice</code> converts <code>DataIntegrityViolationException</code> into a styled 409 page instead of a stack trace.
Form-binding for the <code>Book</code> → <code>Author</code> foreign key	Bound <code>authorId</code> as a separate request parameter; the service fetches the <code>Author</code> and assigns the relationship before save.
Mockito failed on Java 25 (<code>Byte Buddy could not instrument...</code> , Java 25)	Bumped <code>byte-buddy</code> to 1.15.11 and <code>mockito</code> to 5.14.2 in <code>pom.xml</code> , plus added —

Challenge	Resolution
(69) is not supported)	Dnet.bytebuddy.experimental=true to the surefire argLine so the JVM-25 class-file format is accepted during mock generation.

9. Submission Checklist (Rubric Cross-Walk)

Rubric	Where to look
Entities & relationships (10%)	entity/Author.java, entity/Book.java
CRUD functionality (30%)	Controllers under controller/, JSPs under WEB-INF/views/
Inner-join custom query	BookRepository.findAllAuthorsWithBooks + /books/joined page
Spring Boot layering (30%)	controller/, service/, repository/ packages
UI (10%)	webapp/resources/css/styles.css + JSPs
Testing & validation (10%)	src/test/java/... and GlobalExceptionHandler
Documentation (10%)	This file + README.md

10. Figures Index

Figure	Topic	Section
1	H2 console — 10 authors + 10 books seeded	4.1
2	New Author form	4.2
3	Validation errors on form submit	4.2
4	409 page on duplicate email	4.2
5	Authors list	4.3
6	Books list	4.3
7	Inner-join projection page	4.3
8	Inner-join filtered by genre	4.3
9	Edit Author form pre-filled	4.4
9b	Update success — flash + country changed	4.4
10	Dashboard hero with stat cards	5
11	mvn test output — 14/14 tests pass, BUILD SUCCESS	6

End of report.